



Society of Petroleum Engineers

SPE-229013-MS

Qualification and Deployment of a Novel Frac Sleeve System for Extended Reach Wells in Canada's HPHT Duvernay and Montney Formations: A Joint KEC/AU Progression Case Study

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229013-MS](https://doi.org/10.2118/229013-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Kiwetinohk Energy Corp. (KEC), a Canadian energy producer operating in some of the country's most technically demanding unconventional reservoirs, partnered with Advanced Upstream (AU) to trial and deploy a novel, unlimited frac sleeve completion system. The project was initiated to address the persistent operational risks encountered when applying conventional plug-and-perf completion techniques in Canada's Duvernay and Montney formations, with the Duvernay being characterized as a high-pressure, high-temperature (HPHT) environment. Duvernay presents challenges associated with extreme bottomhole pressures, ultra-long laterals, and an elevated risk of casing deformation. In contrast, Montney has pushed the limits of exceptionally high frac slurry rates, heightening the risk of sleeve erosion during stimulation. Together, these factors significantly increase completion complexity, operational risk, and overall cost

Conventional plug-and-perf methods, while widely adopted in unconventional plays, become increasingly less effective as lateral lengths exceed coiled tubing (CT) operational limits and as casing integrity is threatened by seismic activity or high-pressure stimulation events. In such cases, operators risk partial or complete wellbore loss, reduced stimulation effectiveness, and diminished recovery factors.

This paper presents a comprehensive qualification-to-deployment workflow for the AU Sleeve system. The qualification process incorporated an iterative engineering design review, rigorous surface validation, and multiple field-scale trials, culminating in novel in-situ pressure and real-time erosion testing. These efforts aimed to determine the optimal fluid velocity window that ensures reliable stage isolation under the extreme conditions present in KEC's operational areas.

The AU system demonstrated exceptional performance in both laboratory and field environments, meeting stringent mechanical and operational criteria. Following the successful validation phase, the system has been installed and stimulated in four Duvernay and two Montney wells. Results indicate that the technology can maintain effective stage isolation and withstand the erosive forces of high-rate fracturing operations without compromising sleeve functionality.

Montney deployments introduced further complexity due to extremely high frac rates and slurry velocities, conditions that are known to accelerate erosion and reduce the reliability of conventional sleeves.

To address this, a full-scale surface erosion simulation using a complete frac spread was executed prior to field deployment. The outcome was a 100% success rate in sleeve activation and isolation integrity, enabling KEC to optimize stimulation designs for the most demanding operating scenarios.

Introduction

Over the last decade, the unconventional resource development industry has increasingly shifted toward extended-reach lateral designs that routinely exceed 20,000 ft measured depth. This evolution has been driven by the pursuit of greater reservoir contact, improved drainage efficiency, and higher ultimate recovery per well. However, these extended-reach laterals (ERLs) bring with them a host of new engineering challenges, particularly in well completions design, execution, and post-stimulation intervention.

In the Duvernay formation of western Canada — a deep, seismically active, over-pressured play with bottomhole temperatures averaging 250°F and pressures exceeding 13,000 psi — these challenges are compounded. Conventional completion methods, such as plug-and-perf followed by CT millout, are being pushed to their mechanical and economic limits. The high mechanical drag forces associated with ERLs restrict CT reach and increase the likelihood of incomplete plug millouts, while the heightened risk of casing deformation — exacerbated by seismicity — can render sections of the wellbore inaccessible, leading to substantial production and capital losses.

Historically, operators in the Duvernay have mitigated some of these risks by staging operations conservatively or accepting reduced stage counts, but such compromises come at the expense of well productivity and capital efficiency. Plug-and-perf systems also require large volumes of water and proppant to be pumped in fewer, higher-rate stages, which can contribute to greater near-wellbore damage, uneven fracture propagation, and ultimately lower recovery efficiency.

Recognizing these limitations, KEC partnered with AU to investigate a fundamentally different completion approach where the technology offers unlimited stage counts, continuous pumping operations, and eliminates the need for post-frac CT interventions. Moreover, it is engineered to function beyond the reach of conventional intervention equipment, making it ideally suited for the ultralong laterals being deployed in the Duvernay.

Over a two-year development and qualification program, the system was subjected to an exhaustive engineering validation process, including mechanical load testing, erosion resistance assessment, full-scale frac simulation, and multiple field trials. The objective was to confirm that the technology could reliably perform in HPHT environments, withstand the severe erosive conditions present during high-rate fracturing, and maintain functional integrity in the event of casing deformation.

This paper outlines the complete journey from concept validation to successful deployment, providing detailed engineering, operational, and performance data from the Duvernay and Montney applications.

Determining Engineering Requirements

Table 1—Engineering and Field Requirements for Dart-Activated Sleeve Systems

Requirements	Info	Comments
Run through Casing Deformation	Reduced outer diameter (OD) of Dart before activation	This was the highest priority due to the significant risk of casing ovality or collapse during high-pressure stimulation. The system needed to pass through deformed casing to ensure full lateral access even in compromised wellbores.
Dart activation pressure	20,000psi engineered dart	WellCat modeling and actual fracturing data clearly demonstrate that bottomhole treating pressures (BHTP) in KEC's Duvernay Formation can reach as high as 19,000 psi under worst-case conditions. A 20,000-psi activation threshold provides a margin of safety to prevent premature activation.
System pressure rating	Meet or exceed KEC's requirements	Internal pressure yield, collapse pressure, body tensile strength, and torque yield all met KEC's well construction requirements.
Post Screen out options	Rapid recovery capability	In the event of a screen-out, the system had to be capable of fast operational restart to minimize downtime and avoid costly delays.
Casing Drift	>95% of casing ID	Any new completions system considered as an alternative to plug-and-perf must maintain a comparable internal diameter, as significant restrictions would increase friction pressures during fracture stimulation. Additionally, reduced internal diameter could impose limitations on future well interventions
Dissolvability on Flowback	Darts are degradable	Since many sleeves are placed beyond CT reach, dissolvable components are critical to ensuring unobstructed production flowback and future wellbore access.
Field proven diagnostic capability	Compatible with diagnostic imaging tools	The system needed to allow post-installation evaluation via tracers, fiber optics, or other diagnostic tools without impairment.
Multiple Cluster Capability	Single to multiple clusters per stage	Supports both limited entry and conventional cluster designs, offering flexibility in stimulation strategy.
Quality Control	Manufactured to ISO 9001-2015 standards	Ensures repeatable performance through stringent manufacturing and inspection processes.
Perforation Quality	Carbide and carburized inserts available	Allows tailored nozzle and port design to match fracture initiation requirements.
Risk Management	Comprehensive failure mode analysis	Both qualitative and quantitative risk assessments were performed to address potential operational and mechanical failure scenarios before field deployment.

By ensuring that all parameters were met or exceeded during system design and qualification, system was positioned as a technically and operationally viable alternative to plug-and-perf for ERLs in Canada's HPHT unconventional plays. The system was compared to others in market and

Selection Criteria

Multiple unlimited sleeve systems were compared as an alternative to plug and perf in long laterals. Summary of the key selection criteria has been summarized in [Table 2](#) below

Table 2—Section Criteria of Sleeve Systems

Info	AU - Technology 1	Technology 2	Technology 3	Technology 4	Technology 5
Drift % for casing	95%	98%	93.50%	Early stages of development	Tried by KEC Completions Engineers 5 years ago. Poor results plus vendor not focussing on it development
OD when launched	Reduced when deployed - increase when activated - ideal for casing deformation	No reduced OD	No reduced OD		
System	Electronic	Mechanical	Electronic		
Multiple clusters	Yes	Yes (little experience)	Yes (little experience)		
Diagnostic runs with Darkvision	Yes	No	No		
Experience with 5.5"/5.0"	Yes (2 wells by Conoco in Canada)	Yes, but not multiple stages	Yes, but not multiple stages		
QC	Programmed at shop, verified at shop by engineer and verified at location	Similar	Similar		

The intelligent dart is made from a proprietary degradable alloy. It is programmed and launched at the surface to land in a specific sleeve that is to be treated. Reliable, miniature electronic onboard sensors allow the dart to track its location within the wellbore and activate before landing on the target sleeve. Upon pressurizing the completion string, the dart creates differential pressure across the seat and shifts open the sleeve to the zone that needs to be treated. After the fracturing treatment, the dart dissolves, leaving behind a large bore for production.

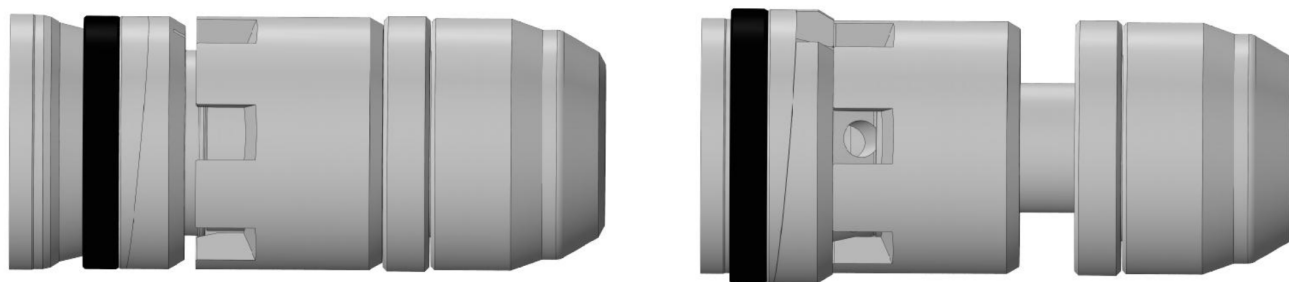


Figure 1—Intelligent Dart non activated (left) and activated state (right)

Progression Roadmap

Following the successful completion of the engineering qualification phase and confirmation that all design criteria were met, a structured progression roadmap was developed for field deployment. This roadmap was designed to progressively increase operational complexity and ensure the system's robustness across a variety of completion scenarios. It included:

- **Laboratory validation** - confirming pressure ratings, mechanical endurance, and erosion resistance under controlled conditions.
- **Surface full-scale frac simulation** - replicating high-rate, high-slurry-volume field conditions in a controlled environment to test activation reliability and erosion thresholds.
- **Initial field pilot wells** - deploying partial systems in representative Duvernay and Montney wells to validate downhole performance and obtain early-stage operational feedback.
- **Expanded field implementation** - deploying the system in additional wells with varying lateral lengths, completion designs, and pumping programs to refine operational best practices.

This systematic approach ensured that each step was informed by data gathered from the previous stage, enabling continuous optimization before large-scale deployment.

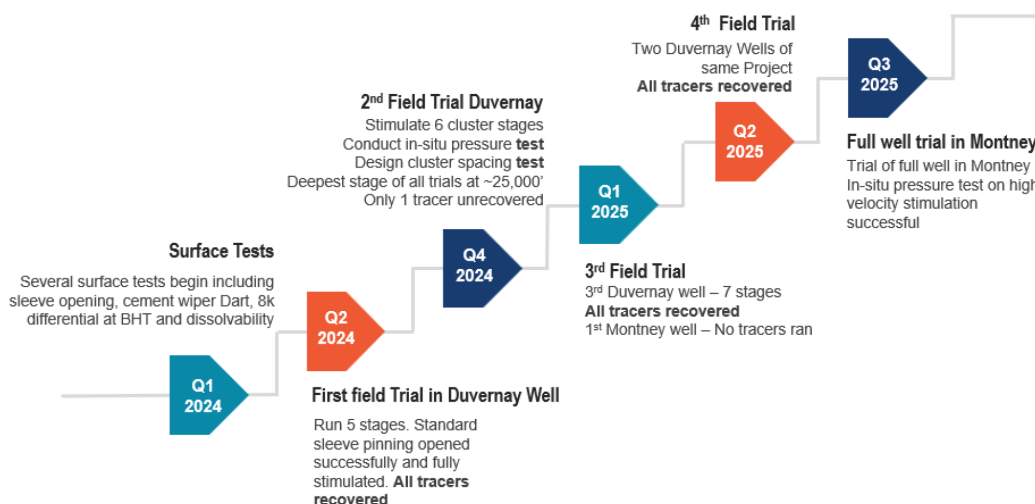


Figure 2—AU system qualification and execution roadmap

Surface Testing

Surface testing played a critical role in validating the Sleeve system before full-scale field deployment. A dedicated surface test facility was configured to replicate real-world frac conditions, including pump rates, slurry concentrations, and fluid properties representative of Duvernay and Montney operations.

The objectives of surface testing were to:

- Confirm the activation reliability of the Darts at specified pressure thresholds.
- Assess erosion resistance of the sleeve ports under sustained high-velocity slurry flow.
- Validate sealing integrity for stage isolation at maximum anticipated pressure differentials.
- Ensure operational compatibility with standard fracturing spread equipment.

The surface tests demonstrated 100% activation success across all tested configurations, with no measurable degradation in sleeve integrity. Port erosion rates were found to be within acceptable limits for long-term stage isolation, confirming the suitability of the sleeve for high-rate slickwater fracturing.

Cluster Sleeve Opening Test using Intelligent dart

Evaluation of Cluster Sleeve Opening and Dart Passage Using Intelligent Dart

Validating the operational effectiveness of the intelligent dart's ability to shift open multiple sleeve types is essential for qualifying AU systems. The test was designed to confirm that the activated intelligent dart can successfully count Single Point (SP) and shift and open the Cluster Sleeves (CS), as well as a Landing Sleeve (LS) at the end of the target zone.

The test system consisted of three cluster sleeves (CS1, CS2, CS3), two single point sleeves (SP1, SP2), and one landing sleeve (LS). All components were connected in series with 5.5" 23# casing. Supporting equipment included a pump truck, hydraulic choke valve, pressure relief valve, treating iron, and connecting hoses. The intelligent dart was loaded at the upstream end and pumped through the assembly using water at a rate of 600 L/min.

The downstream end of the test jig incorporated an adjustable hydraulic choke valve, set to maintain a backpressure of 2000 psi within the casing line, simulating typical downhole fluid conditions.

Test Procedure. The intelligent dart was programmed to simulate the actuation sequence for a six-stage completion. The test involved two separate pumping cycles through the assembly:

- **Pumping Cycle 1:** During the first cycle, the dart was pumped through the test string, passing through all sleeves without activation. The dart's internal electronics recorded the passage through each sleeve type, confirming recognition of the SP and LS sleeves. No significant pressure changes were recorded, and the dart was recovered downstream in its original form, indicating successful non-activation transit.
- **Pumping Cycle 2:** In the second cycle, the dart was again pumped at a rate of 600 L/min from the upstream end. This cycle was programmed for the dart to recognize SP1 and SP2 as Stage 3 and Stage 2, respectively, and then activate to open the designated cluster sleeves of Stage 1. Upon reaching cluster sleeve CS1, the dart landed on the sleeve's specific profile, which is designed to isolate pressure and allow pressure build-up. This build-up enabled the ports of the cluster sleeve to shift and open. The profile then expanded to allow the fully activated dart to pass through the sleeve and continue to the cluster sleeves CS2 and CS3. After successfully opening all three cluster sleeves, the dart traveled through CS3, landed on the landing sleeve (LS), shifted it open, and remained seated to provide pressure isolation below the test line.

Observations.

- **Cycle 1:** The dart passed through all sleeves without incident. There were no notable fluctuations in pressure readings, and physical inspection confirmed that the dart remained undamaged at the end of the test string.
- **Cycle 2:** The activated dart successfully shifted open each of the targeted cluster sleeves (CS1, CS2, CS3) and then landed and opened the landing sleeve (LS). The presence of water jets from the open sleeves confirmed successful shifting. The dart effectively provided pressure isolation after seating in the landing sleeve.

Conclusion. The intelligent dart demonstrated reliable performance by passing through non-targeted sleeves, accurately identifying each sleeve type, activating as programmed, and successfully shifting open all targeted cluster and landing sleeves. The dart maintained structural integrity throughout both cycles and effectively provided pressure isolation in the final seat. These results validate the intelligent dart for controlled multi-stage sleeve opening in fracturing operations.

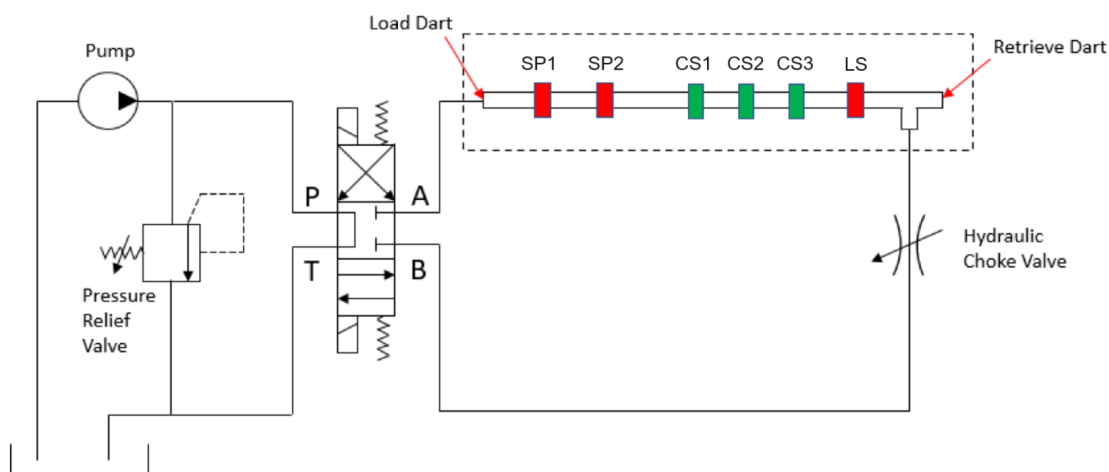


Figure 3—Dart Functional Test Schematic with Cluster Sleeves

Wiper Dart flow test

Demonstrating that cementing wiper darts can be pumped through sleeve assemblies without causing accidental actuation is essential for the reliability of running AU frac sleeves. To verify this, a series of tests were conducted by passing wiper darts through the AU sleeves with careful monitoring of system pressure and sleeve integrity.

Test Apparatus and Setup. The assembled test string comprised a cluster sleeve and a landing sleeve connected to a surface launcher through a 16-foot section of 5-1/2" tubing. Both sleeves were pinned with brass shear screws rated at 315psi per screw, as per field specifications, to allow controlled opening in response to significant differential pressure. A triplex pump was connected at the upstream end to simulate operational flow conditions. Pressure and flow transducers were installed and linked to a data acquisition system (DAS) to enable real-time monitoring of all test parameters.

Test Procedure.

- The landing sleeve was first drifted with AU sleeve specific drift to verify full bore access, and standard shear screws were installed.
- The test string was assembled and checked for leaks under pressure.
- All instrumentation was connected and the DAS was initialized for continuous data recording.
- A cement wiper dart was loaded into the launcher, and the system was brought up to a flow rate of 2 bbl/min
- The dart was pumped through the assembled sleeves, with pressure readings and sleeve positions monitored throughout.
- This process was repeated at increased flow rates of 3.7bbl/min and 6.3bbl/min
- After each test sequence, the sleeves were visually and mechanically inspected to verify that no unintentional movement or damage had occurred.

Results and Observations. At all tested flow rates, the cementing wiper dart successfully passed through both the cluster and landing sleeves without inducing any pressure spikes or sleeve movement. The system maintained stable pressures, and post-test inspection confirmed that the sleeves remained in their original positions, with no evidence of accidental actuation or abnormal wear on the wiper dart. These results confirm that the selected cementing wiper dart is mechanically compatible with the frac sleeve assembly and suitable for use in operations without risk of unintended sleeve activation.

Dart differential testing

In order for KEC to adopt AU's technology an extensive qualification test program was conducted to verify the ability of the degradable alloy dart to hold high differential pressures and maintain sealing integrity throughout multi-cycle and extended duration testing.

Test Procedure.

- The activated degradable alloy dart was positioned in a test sleeve engineered with an internal profile identical to the field landing frac sleeve.
- The dart was manually inserted and seated, after which hydraulic pressure was applied using a Haskell pump system to fully energize the elastomeric seal elements.
- The test apparatus, including both uphole and downhole sections, was filled completely with fresh water.
- The temperature within the test fixture was elevated to 248°F using a controlled heating system, with insulation to ensure uniform heat distribution.

- With the fixture at temperature, pressure was increased on the uphole side to 16,000 psi. The downhole side was then pressurized to 8,000 psi, establishing an initial differential of 8,000 psi; this was held for 15 minutes. Subsequently, the uphole pressure was reduced to 10,000 psi and maintained for 5 minutes, resulting in a 2,000psi differential. The complete cycle test order is shown in the test flow schematic.
- The cycling sequence was repeated: the uphole side was again pressurized to 16,000 psi and the downhole side to 8,000 psi, held for 5 minutes, followed by another bleed down to 10,000 psi uphole.
- The differential pressure was brought to 8000psi again and held for 5 minutes before being reduced to 200psi differential. A minimum pressure differential of ~200 psi was maintained throughout these cycles to prevent unseating of the dart spigot.
- This cycle was repeated twice, each time holding 8,000 psi differential for a shortened period, then bleeding down as described.
- For the final sequence, the dart was subjected to 8,000 psi differential for a continuous three-hour duration to validate extended seal integrity under maximum stress conditions.
- Throughout all stages, pressures were continuously monitored on both uphole and downhole sides using calibrated sensors. Any variations observed during the dissolution phase were also recorded, with additional pressure adjustments made as required to maintain test conditions.
- Upon completion of the pressure cycles, the pressure was safely bled off and the dart was removed and inspected for physical condition, seal integrity, and any signs of material degradation.

Results and Observations. The degradable alloy dart successfully maintained pressure isolation throughout all cycling and extended duration tests, meeting or exceeding KEC's qualification requirements. No pressure loss, leak path formation, or premature seal degradation was observed during any phase of the test program. Post-test inspection of the dart confirmed intact elastomeric seals and minimal physical wear, supporting the suitability of the dart design for high-pressure, multi-cycle fracturing applications.



Figure 4—Surface differential test apparatus

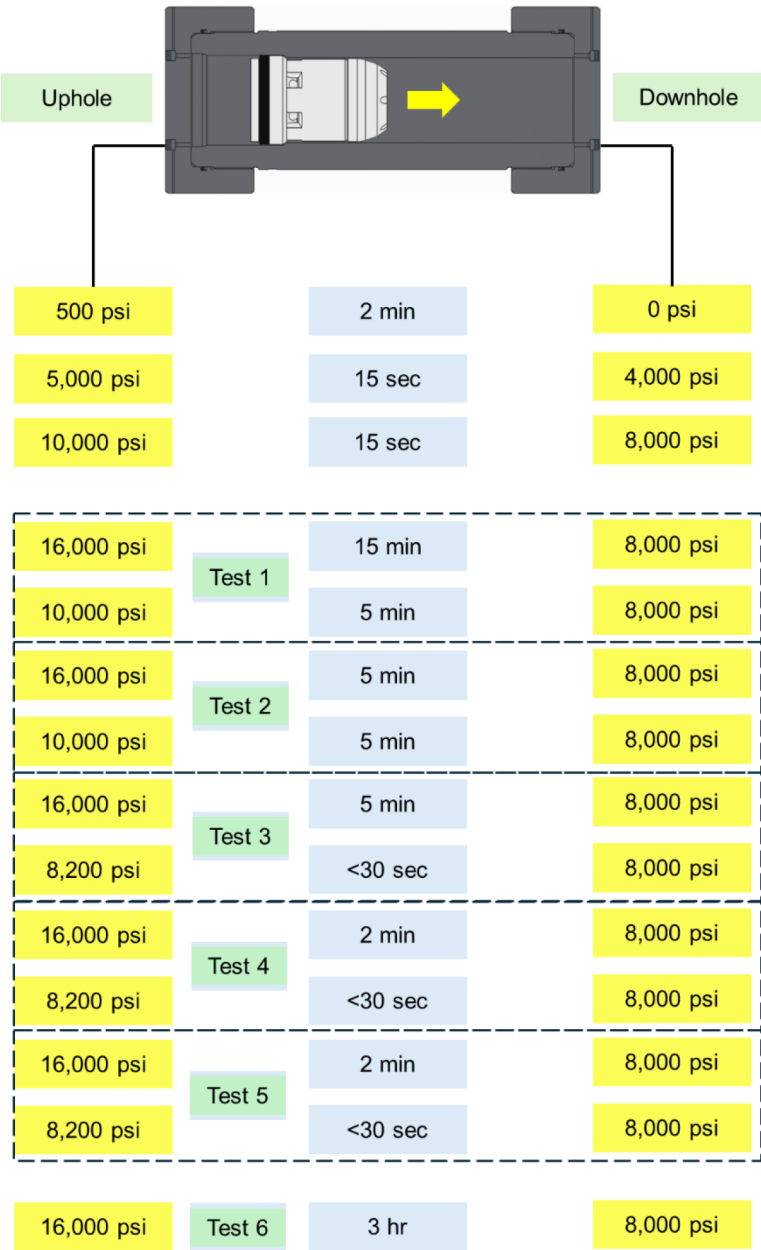


Figure 5—Differential Cycle Test order

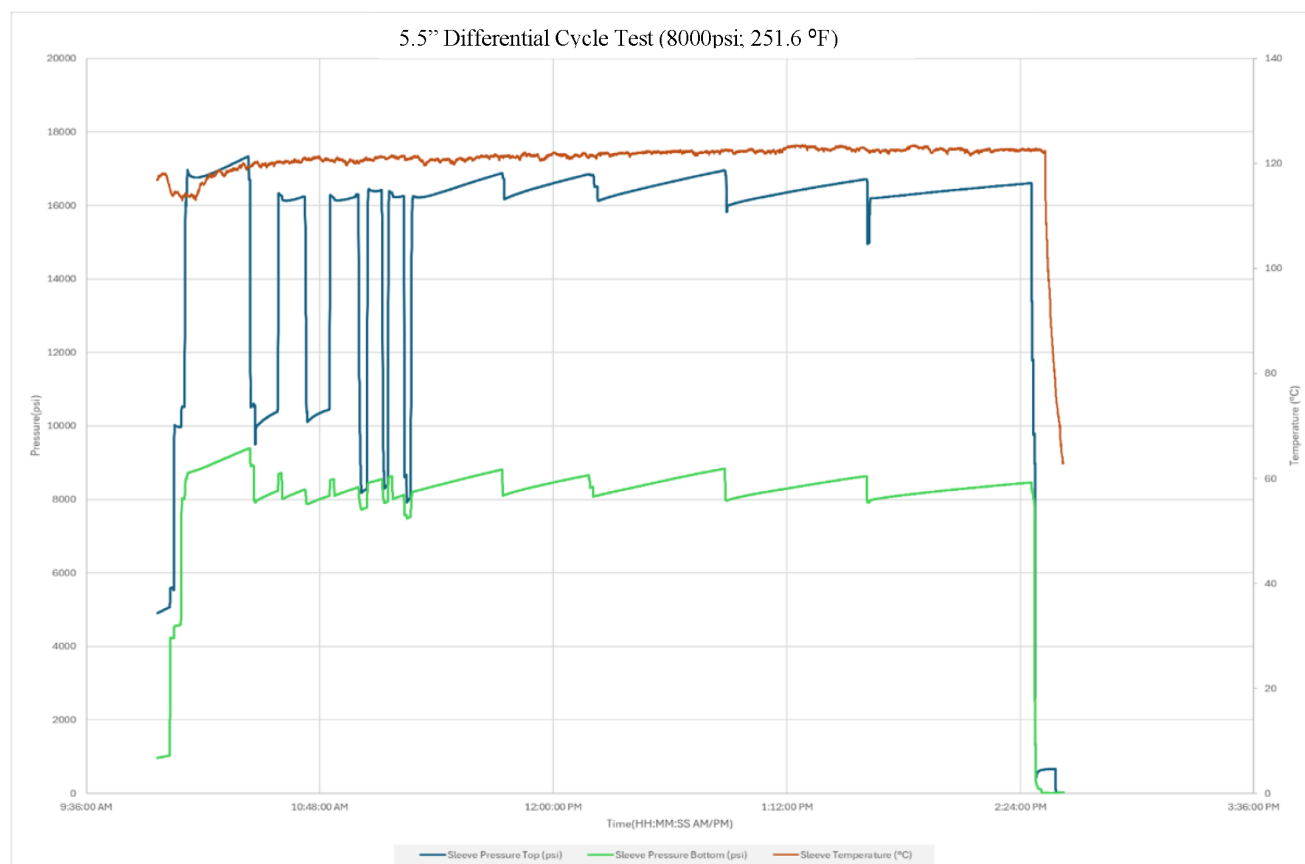


Figure 6—Differential Test Chart

Dissolution Test

A key advantage of using AU systems is their ability to bring wells into production without the need for coil tubing cleanout runs, making them particularly suitable for extended reach applications. This is achieved through the dissolution of the AU intelligent dart in well fluids, which eliminates residual downhole obstructions. It is important to test the dissolution capability of these darts under realistic fluid conditions to ensure the selection of appropriate degradable materials and maintain reliable wellbore access after fracturing operations. A controlled laboratory test was conducted at elevated temperature using a 50:50 mix of freshwater and produced water. For this test, a temperature of 203°F was selected to avoid uncontrolled exothermic temperature increases, even though this is lower than the actual reservoir temperatures found in the Duvernay wells. As dissolution rates increase with temperature, it is expected that field dissolution will occur more rapidly than observed in the laboratory results.

Test Procedure.

- The dissolution vessel was filled with a 50:50 mixture of produced water and freshwater to replicate downhole salinity and composition.
- The intelligent dart was placed inside the chamber, and both ends were secured. A vent hose was connected to the chamber to allow safe release of hydrogen gas generated during the dissolution reaction.
- The vessel was wrapped with heat tape, and the heating process was initiated. The dissolution test duration commenced once the vessel temperature reached 203°F.
- Fluid levels in the vessel were periodically topped up with the same 50:50 water mixture to compensate for consumption during the dissolution process and ensure the dart is fully immersed in fluid. This step was maintained throughout the duration of the test.

After six days of continuous exposure at the set temperature, heating was discontinued. The dart was removed from the vessel, thoroughly rinsed, and detailed measurements of mass and remaining dimensions were recorded.

Results and Observations

The degradable dart maintained pressure integrity during the initial differential pressure test at 203°F. Over the six-day dissolution period, the dart exhibited a mass loss of approximately 76%, confirming the effectiveness of the dissolution process under these test conditions. Post-test inspection revealed that only small fragments of non-dissolvable materials remained. The majority of the dissolvable alloy had broken down, and in actual downhole applications, it is expected that any residual material would further fragment and be transported with production fluids.

The extended six-day test at 203°F produced promising results and, given that many downhole environments exhibit even higher temperatures, an accelerated dissolution rate can be anticipated under field conditions.



Figure 7—Dart degraded in the test apparatus after 6 days of dissolution

Well Stimulation Testing

Well stimulation trials were particularly focused on the success of Dart landing/isolating, dissolvability (Tracer results), in-situ pressure test and post frac-screenout recovery

The following figure displays a typical AU system stage deployed in Duvernay formation. Dart can be seen successfully landing, shifting and isolating from previous stage



Figure 8—Treatment chart of an AU cluster stage in Duvernay

To quantify downhole performance under actual reservoir conditions, the system was deployed in both Duvernay and Montney wells with varying measured depths (MD) and lateral lengths (LL). Tracers were incorporated into the stimulation program to evaluate dissolvability and flow assurance

Table 3—Summary of Water Tracer Results

Project	Formation	MD (ft)	LL (ft)	Tracers Pumped	Tracers Recovered
1	Duvernay	23,320	10,274	2	2
2	Duvernay	26,483	13,343	3	2
3	Duvernay	24,599	11,515	4	4
	Montney	22,113	12,353	0	N/A
4	Duvernay	24,532	11,450	3	3
5	Duvernay	25,574	15,711	3	2
	Montney	21,363	8,375	6	TBD

In-situ pressure test

To validate sleeve sealing performance in live well conditions, an in-situ pressure test was conducted on a Montney well. The test involved applying a sustained differential pressure of 8,000 psi across an isolated sleeve for a duration of 90 minutes.

Key observations:

- No measurable pressure decay was recorded during the test interval.
- Sleeve sealing elements maintained full isolation integrity.
- Flowback recovery to operational readiness was achieved in less than one hour.

In-line Field Erosion Testing

As part of the qualification process, a surface erosion test apparatus was deployed on project 4, directly into the frac trunk line feeding the wells on the pad. The surface test assembly consists of a nozzle that reduces flow from a 7-1/16" spool to a 4-1/2" casing tail joint, to which the AU single point sleeve with the erosion resistant coating is attached. This assembly is installed within a 10-foot length of trunk spool with a 7-1/16" diameter and is positioned after wells A and B but before well C, so it only sees the proppant that is dedicated to well C. This setup ensures that all proppant directed to well C passes through the sleeve for erosion evaluation. The test sleeve was an AU single point sleeve with Erosion Resistant Coating-1 (ERC-1) on the inner sleeve. This in-line configuration provided direct, real-time measurement and evaluation of erosion performance in an environment closely replicating actual field service conditions.

The test assembly was subjected to an erosional load totaling 14.77 million pounds of 40/70 Northern White sand, pumped at an average slurry rate of 94.3 bbl/min, with peak rates reaching 119bbl/min. After completion of the surface erosion test, the sleeve was disassembled at the AU facility for inspection. Both visual examination and dimensional measurements were performed to assess erosion. Subsequently, an AU intelligent dart was installed in the sleeve, and a differential pressure test was conducted at 8,000 psi. The test confirmed pressure integrity, with only minor surface wear observed on the coated seat and a slight change in the angle of the sleeve profile.

A second qualification test was conducted using the same model of AU single point sleeve, this time featuring Erosion Resistant Coating- 2 (ERC-2) on the inner sleeve. The sleeve was exposed to peak flow rates of approximately 119 bbl/min and a total proppant load of 28.66 million pounds of 40/70 Northern White sand. Following the surface erosion test, the sleeve was disassembled and inspected at the AU facility, with comprehensive visual and dimensional measurements performed. An AU intelligent dart was then installed in the sleeve, and a differential pressure test at 8,000 psi was carried out. The dart successfully maintained pressure isolation for the entire three-hour duration of the test. Results demonstrated further improvements in erosion resistance and consistent pressure integrity, with measurements indicating significantly less damage to the inner sleeve compared to ERC-1.

Results

The qualification program, which included both controlled laboratory testing and surface in-line field trials, clearly demonstrates that increasing the thickness of the erosion resistant coating on inner sleeve surfaces significantly enhances durability for high-rate hydraulic fracturing. Sleeves protected by these coatings consistently maintained pressure integrity and structural performance even after exposure to demanding, sand volumes and slurry velocities. The successful results of the erosion tests provided strong confidence in the effectiveness of the erosion resistant coating, enabling the design and execution of fracturing treatments at higher pump rates while minimizing the time required to treat each stage for a given proppant volume. As a direct result of this success, KEC was able to deploy AU sleeves with this coating along the entire wellbore—covering over 50 stages per well—and efficiently frac and treat every stage at improved operational efficiency, ultimately reducing completion costs per pad. This strategy addresses the known limitations of nitride coatings and enables robust seat performance with reliable isolation, even during the most demanding fracturing operations in the Montney and similar unconventional reservoirs.

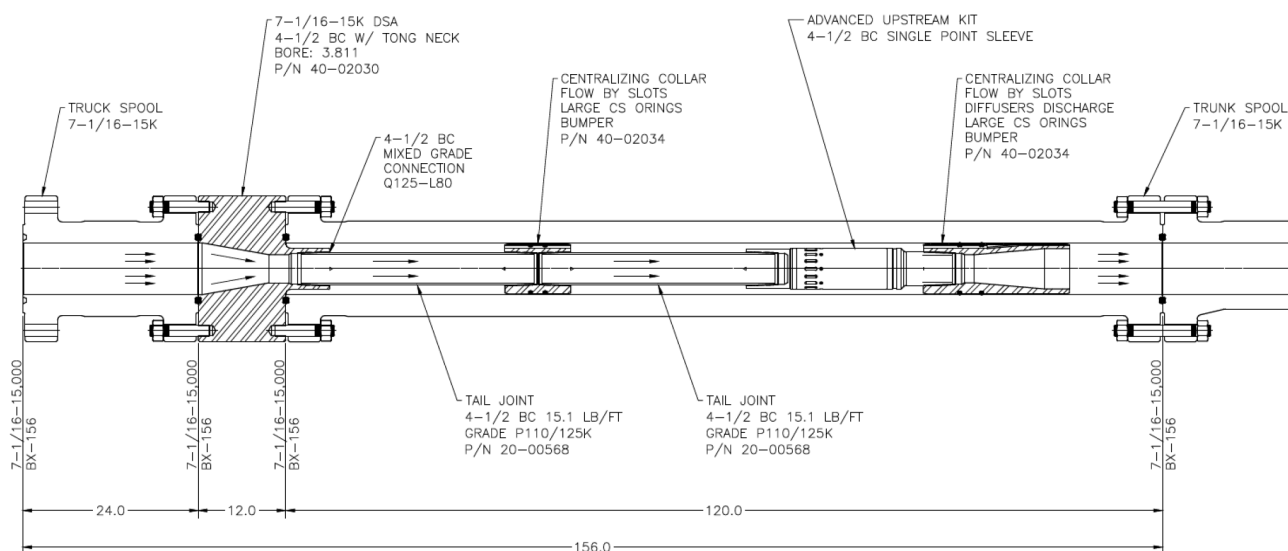


Figure 10—Erosion test setup layout drawing



Figure 11—Surface erosion test gear in line with frac line

Conclusions

The qualification and deployment of the Sleeve system in KEC's Duvernay and Montney wells represent a successful transition from concept to proven field technology. Key takeaways include:

- **Operational Reliability** — The system achieved 100% activation success during both surface and field trials, validating its ability to operate in ERLs beyond CT reach.
- **Stage Isolation Integrity** — In-situ testing and tracer analysis confirmed effective zonal isolation under high-pressure, high-rate stimulation conditions.
- **Erosion Resistance** — Sleeve ports demonstrated excellent wear resistance, even in the high-slurry-velocity environments typical of Montney completions.
- **Operational Efficiency** — Eliminating post-frac CT interventions reduced completion cycle time, improved operational safety, and enhanced overall capital efficiency.

- **Applicability Across Formations** — The system proved effective in both Duvernay HPHT conditions and Montney's high-rate, erosive environments.

References

- Li, N., Zhang, Y., Song, N., and Zhang, H. Numerical Study on Improving the Erosion Life of Ball Seat for Oil and Gas Reservoir Fracturing. *Engineering Failure Analysis* **60**: 188–203. 2015.
- Young, G.E., and Luker, T.C. Erosion Prediction and Control in Oil and Gas Production. OTC-23085-MS, presented at the Offshore Technology Conference, Houston, Texas, 30 April-3 May. 2012.
- API (American Petroleum Institute). Guidelines for Erosion/Corrosion Management in Oil and Gas Production Facilities, API Recommended Practice 14E. Washington, D.C. 1991.